

Same Budget. More Water.

Ranking Egypt’s irrigation budget by **the water that evaporated without growing a crop.**

DECISION-MAKER

MWRI budget committee

MANDATE

Egypt’s Nile allocation

TOOL

IPAT, already at MWRI

WHERE

Nile Delta & Valley

The tool works. The decision still does not get made.

FAO lists more than twenty working WaPOR applications. Egypt has its own, running inside the ministry.

IPAT

Irrigation assessment tool for Egypt. Operational pilot, hosted by the Ministry of Water Resources and Irrigation. WaPOR v3 Level 3 at 20 m, across ~1.5 million hectares of the Middle and West Delta.

It shows, accurately, where water productivity is high and where it is low.

It does not say where to spend.

The decision it does not make

A planning officer with a fixed annual modernisation budget still has to decide **where to spend it**. Today that split is made by cultivated area or political balance across governorates.

Nothing in the toolchain says which hectare of subsidy buys the most water back.

The gap is not data, and it is not accuracy.

The output arrives in the wrong units for the decision: millimetres per dekad, when the decision is denominated in pounds, hectares and cubic metres.

WaPOR knows which water grew something

It reports evaporation, transpiration and interception as separate layers. Nobody converts that into money.

TRANSPIRATION

passed through a stoma and grew a crop

77%

EVAPORATION

delivered, paid for, grew nothing

23%

Study-area average share of all water consumed, measured over 18 dekads. The orange share runs from 0.16 to 0.29 across governorates; the volume behind it varies far more.

4,844

Mm³ evaporating each season across the study area, water already delivered and already paid for

Recoverable

with measures that already exist in Egypt: laser levelling, mulching, better rice water management

Everywhere

no new instruments, no new field campaign, the data exists today for every WaPOR country

So the question of where the money should go has a measured answer: **wherever a pound of spending recovers the most evaporated water.**

Recoverable evaporation is an upper bound: some avoided evaporation becomes transpiration and yield rather than saved water.

The mandate, the gap in it, and why this tool fills it

Who makes the decision, what they are deciding, and where this tool already sits.

THE MANDATE

Ministry of Water Resources and Irrigation

Allocates and manages Egypt's Nile share under the National Water Resources Plan, and is accountable to the Ministry of Finance for what the on-farm modernisation budget delivers. Its remit is explicitly to raise water-use efficiency. Egypt has roughly 55 BCM a year and around 110 million people.

The decision-maker is the annual budget committee, working on an annual cycle, under pressure to spread allocations evenly across governorates.

THE CHALLENGE

A mandate to save water, no evidence on where

The committee must decide which governorates receive this season's budget, and in what proportion. Today that split can only be defended by cultivated area or by political balance, because nothing tells it where a pound of subsidy recovers the most water.

The gap is not data, and it is not accuracy. The output arrives in millimetres per dekad; the decision is denominated in pounds, feddans and cubic metres.

THE ALIGNMENT

IPAT already sits inside the mandate

IPAT is operational at MWRI on WaPOR v3 Level 3 across roughly 1.5 Mha of the Middle and West Delta. It shows, accurately, where water productivity is high and low, and stops there.

We add the one step that converts its output into the committee's decision: rank by recoverable evaporation, price it per cubic metre, and spend down the list. No new tool, no new institution, no new meeting.

In practice: *the planning officer works down a ranked list instead of dividing the budget by cultivated area.*

The method is three lines

Deliberately so. Everything it depends on is visible and editable.

$$\begin{array}{ll} \text{recoverable water} & \text{m}^3 \text{ per year} \\ V & E_{\text{mm}} \times \text{irrigated ha} \times 10 \times \text{reduction} \times \\ & \text{seasons} \end{array}$$

$$\begin{array}{ll} \text{annualised cost} & \text{USD per year} \\ C & (\text{USD per ha} \div \text{life years}) \times \text{irrigated ha} \end{array}$$

$$\begin{array}{ll} \text{cost of water} & \text{USD per m}^3 \\ C & \div V \end{array}$$

Then sort by c and spend down the list.

Cheapest water first. A district can be part-treated, so the greedy solution is optimal.

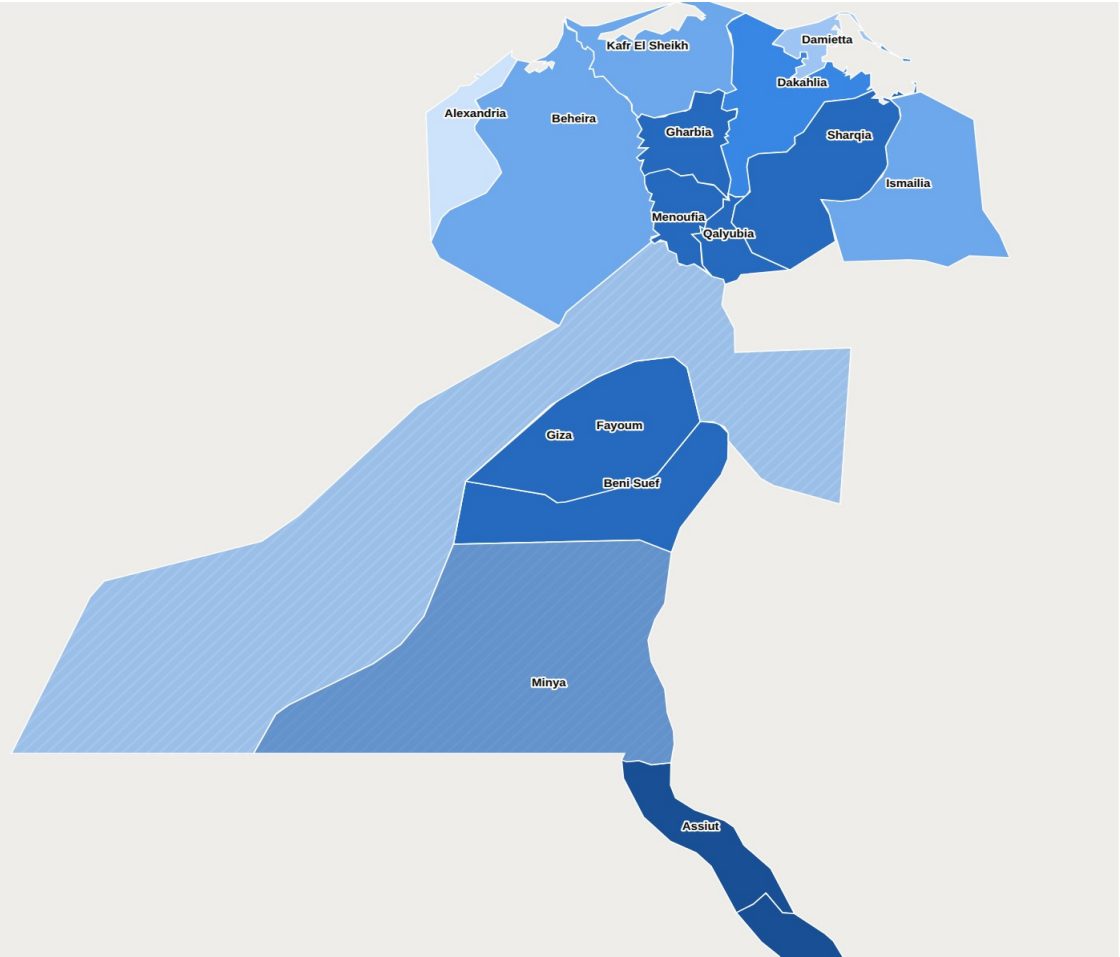
Compared against an area-proportional split, a neutral prior, not a claim about ministry practice.

The cropland mask is free

Outside the irrigated Nile there is no water and effectively no rain, so seasonal evapotranspiration collapses to near zero. A seasonal AETI threshold separates cropland from desert almost perfectly, using data we already have, with no extra product, licence or cost. Open water is removed separately, by beneficial fraction and by JRC surface water.

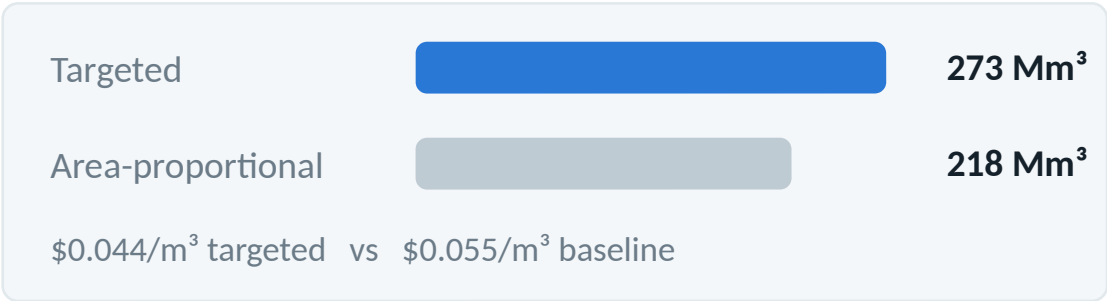
Same budget, same measures, different order

At a \$12M modernisation budget



+25%

more water recovered than an area-proportional split, and the same figure on unit cost, so it is not an artefact of spending more



Evaporation share runs 0.16 to 0.34 across the study area, a twofold spread, and the reason targeting works at all. The ranking is driven by that measurement; the size of the prize still moves with every cost coefficient.

FAO WaPOR v3 Level 2, 100 m, 18 dekads, 1 May – 31 Oct 2024. Observed irrigated area 3.56 Mha against 3.21 Mha reported cultivated, an independent check the method passes.

What the data told us we had wrong

We expected the northern Delta rice belt to come out worst. Ponded paddy, evaporating all season. The measurement says otherwise.

share of consumed water that evaporated without growing anything · $E / AETI$ · area-weighted

Delta, rice belt

what we predicted would be WORST



Nile Valley (Upper Egypt)

684 mm through the field



Delta, other

500 mm through the field



A closed rice canopy shades the water. Ponding does not imply evaporation, Damietta reaches 0.157 and Kafr El Sheikh 0.191, the most efficient irrigated land we measured anywhere in Egypt. And the second surprise: the wasted **fraction** is similar everywhere, around 0.24. What differs is **throughput**, Upper Egypt is hotter, pushing 660–775 mm through a field against the Delta's 510–640, so the same inefficiency costs more water. Targeting follows the heat, not the bad practice.

A recommendation a committee can sign

Pure cheapest-first funds seven governorates out of eighteen. No ministry with an equity mandate signs that.

No floor

7

governorates funded

273 Mm³

maximum water

25% floor

18

governorates funded

264 Mm³

-3.2% vs pure targeting

50% floor

18

governorates funded

253 Mm³

-7.2% vs pure targeting

100% floor

18

governorates funded

218 Mm³

-20.1% vs pure targeting

Reserve a quarter of the budget for equity and you fund every governorate for 3.2% of the water. Stakeholder engagement happens outside the dataset interface, so we priced the political constraint instead of optimising around it. The floor is a slider: the committee argues with the model in the room and leaves owning the result.

And next season, we check

Most tools stop at the recommendation. This one can be checked against what actually happened.



Why this matters

The same measurement that justified the spending also measures the result, so progress is observed rather than reported. We will publish the confidence interval with it, including the cases where the change is too small for WaPOR to see.

What we are not claiming

The limits of the method, set out plainly.

Field scale is not basin scale

Some water an intervention frees was already returning to a drain and being reused downstream, and suppressing evaporation can convert it to transpiration and yield rather than to saved water. Recoverable E is an upper bound. These figures rank places; they are not a water account.

Feasibility rests on one swap

Governorates are not how this budget is delivered, irrigation directorates and command areas are. The method is boundary-agnostic: the same three lines run on any polygon set, including IPAT's own 20 m command areas. That swap is the whole distance between this and something MWRI could run.

We caught ourselves contaminating our own result

Lake Nasser, the Toshka lakes, canals and fish ponds survived both the JRC water mask and the beneficial-fraction test, evaporating 1,300–1,800 mm. In Aswan that was 5,392 pixels and it had put Aswan top of the national ranking on contamination alone. A physical ceiling on cropland evaporation removed it; Aswan's lead largely went with it.

Most of the value is the pattern, not the pixel

Real per-district evaporation gives +25.2% over the baseline. A per-system average alone gives +21.5%. So satellite detail is worth about 3.7 points, the rest is the system-level pattern. But that pattern is precisely what the satellite told us we had backwards.

Each of these is documented in full, with the corrections we made, in the accompanying technical note.

WHY IT WOULD ACTUALLY GET USED

An ordered spending list, not another map.

- **No new tool** it runs on IPAT, already operating inside the ministry
- **No new meeting** the output is the brief the budget committee already writes
- **No new data purchase** WaPOR is free and global, so this is not Egypt-specific
- **No new mandate** raising water-use efficiency is already what MWRI answers for

An existing tool, an existing budget committee, an existing annual cycle.